INVENTORY MANAGEMENT WITH POINT OF SALE SYSTEM USING FEATURED-BASED RECOMMENDATION ALGORITHM FOR FIVE BROTHERS TRADING IN CARMONA, CAVITE

Undergraduate Thesis
Submitted to the Faculty of the
Department of Industrial and Information Technology
Cavite State University – Carmona Campus
Carmona, Cavite

In partial fulfilment
of the requirements for the degree
Bachelor of Science in Computer Science

LESTER JOHN M. AGUSTIN
EDMON B. NOFIES
JOHN MICHAEL C. PAGUIBITAN
ASHLEY ROVELL D. SISON
November 2025

BIOGRAPHICAL DATA

Lester John M. Agustin was born on April 11, 2003, in Silang, Cavite. He is the eldest of two children of Mr. Michael H. Agustin, a printing technician, and Mrs. Haydee M. Agustin, a research and development (R&D) engineer at United Graphic Expression Corporation (UGEC) Technopark. He has one sibling, his younger brother, Dexter Jake M. Agustin. He is a Filipino citizen, Catholic, and currently lives at Block 7 Lot 5, Carissa Homes, Paliparan 1, Dasmariñas, Cavite.

He finished his elementary and secondary education at Mahonri Academy and Science High School from 2010 to 2022. In September 2022, he enrolled at Cavite State University (CvSU) – Carmona Campus to pursue a Bachelor of Science in Computer Science.

Edmon B. Nofies was born on February 11, 2002 in Biñan, Laguna. He is the youngest among the 3 children of the late Mr. Eddie Nofies and Ms. Monica Nofies, a housewife. He is currently residing at Olivarez Homes South Compound Blk 3 Lot 12 Sto. Tomas, Biñan, Laguna.

He obtained his primary education at San Vicente Elementary School Biñan, Laguna from 2008 to 2014. He then completed his secondary education at Biñan Secondary School of Applied Academics, Biñan, Laguna from 2014 to 2018. He pursued his senior high school education at Biñan City Senior High School Sto. Tomas Campus, Biñan, Laguna from 2019 to 2021. In 2022, he enrolled at Cavite State University - Carmona Campus, Carmona, Cavite for his college education in the Bachelor of Science in Computer Science program.

During his college years, he became an active member of YPADS (Young Programmers and Developers Society), a college organization connected to his course. He is currently in his fourth year for the Academic Year 2025-2026 and is diligently working toward completing the requirements for his degree.

John Michael C. Paguibitan was born on October 9, 2002, in Iriga city Camarines sur. He is the eldest son of Mr. Michael C. Paguibitan, a sewing mechanic, and Ms. Jocelyn C. Paguibitan, a sewer at Reliance Producer Cooperative. He has three silbings. Joan Mickaela C. Paguibitan the sister, Yvonne C. Paguibitan, the youngest sister, and Michael C. Paguibitan Jr. The younger brother. He is residing at 12677 Blk 2 Lot 22, Phase 2 Milagrosa, Carmona, Cavite.

He completed his Primary education at Carmona Elementary School in 2012, Secondary education at Carmona National High School in 2016, and Senior High School at Angelo Levardo Loyola Senior High School, Carmona, Cavite in 2022. In the same year, he enrolled at Cavite State University – Carmona Campus.

Ashley Rovell D. Sison was born on August 28, 2004, in Metro Manila. She is the eldest among the two children of Mr. Roberto P. Sison, a horse groomer; and Ms. Shirly D. Sison, a merchandiser. She is currently residing at Blk 16 Lot 34 Humility Street, Mabuhay, Carmona, Cavite.

She obtained her primary education at Mabuhay Elementary School in 2016 and her secondary education at Carmona National High School, Carmona Cavite in 2020 and her senior high school with high honors at Angelo Levardo Loyola Senior High School, Carmona Cavite in 2022.

In the following year, she enrolled at Cavite State University (CVSU) – Carmona Campus for her college education in the Bachelor of Science in Computer Science program.

ACKNOWLEDGEMENT

The proponents would like to express their heartfelt gratitude to the following individuals who generously shared their time, effort, and support in the proposed study:

To Mr. Jhumel C. Ignas, our thesis adviser, for his valuable guidance, constructive suggestions, and continuous encouragement that greatly contributed to the improvement and success of this research.

To Ms. Jackelyn R. Murallon, our technical committee member, for her insightful feedback, expertise, and assistance in refining the technical aspects of our study.

The group expresses their appreciation to Ms. Janine B. Bacosmo, Department Research Coordinator; Dr. Regene G. Hernandez, Department Chair; Dr. Carlo Emil B. Mañabo, Campus Research Coordinator; and Dr. Jenny Beb F. Espineli, Campus Administrator, for their guidance, encouragement, and administrative support throughout the conduct of this research.

Lastly, the proponents extend their deepest appreciation to their families, classmates, and friends for their constant motivation, understanding, and moral support that inspired them to accomplish the proposed study successfully. Above all, they give thanks to Almighty God for the wisdom, guidance, and strength bestowed upon them during this journey.

TABLE OF CONTENTS

Page

BIOGRAPHICAL DATA ii
ACKNOWLEDGEMENT iv
TABLE OF CONTENTS v
LIST OF FIGURES vii
LIST OF APPENDIX FIGURE ix
INTRODUCTION 1
Objective of the Study 2
Significance of the Study 3
Time and Place of the Study 4
Scope and Limitation of the Study 5
Definition of Terms 5
Technical Terms 5
Operational Terms 6
REVIEW OF RELATED LITERATURE 8
Point of Sale Systems 9
Inventory Management System 9
Feature-Based Product Recommendations System 10
Agile Development Method 10
Synthesis Review of the Study 11
METHODOLOGY 12
Material 12
Methods 12
Requirements 14
Design 15
Development 21
Testing 21
Unit testing 21
System testing 21

Deployment. 24
Review. 24
REFERENCES 25
APPENDICES 27

LIST OF FIGURES

Figure **Page**

1 Figure 1. Agile Model (Eby, 2017) 13

2 Figure 2. Input-Process-Output (IPO) diagram of the manual system 15

3 Figure 3. IPO diagram of the Inventory Management with Point of sale System using Featured-Based Recommendation Algorithm for Five Brothers Trading in Carmona, Cavite 16

4 Figure 4. Use Case Diagram of the Development of Inventory Management with Point of sale System using Featured-Based Recommendation Algorithm for Five Brothers Trading in Carmona, Cavite 18

5 Figure 5. Entity Relationship Diagram (ERD) of the Inventory Management with Point of sale System using Featured-Based Recommendation Algorithm for Five Brothers Trading in Carmona, Cavite 19

6 Figure 6. Algorithm Diagram of the Inventory Management with Point of sale System using Featured-Based Recommendation Algorithm for Five Brothers Trading in Carmona, Cavite 20

7 Figure 6. Screenshot of Login Page 40

8 Figure 7. Screenshot of Dashboard 40

9 Figure 8. Screenshot of Category Page 41

10 Figure 9. Screenshot of Products Page 41

11 Figure 10. Screenshot of User Information 42

12 Figure 11. Screenshot of Records Top Selling Page 42

13 Figure 12. Screenshot of Records Sold Items Page 43

14 Figure 13. Screenshot of Records Critical Stocks Page 43

15 Figure 14. Screenshot of Records Cancelled Orders Page 44

16 Figure 15. Screenshot of Category for Cashier Page 44

17 Figure 16. Screenshot of Products for Cashier Page 45

18 Figure 17. Screenshot of Point of Sale Part 1 for Cashier Page 45

19 Figure 18. Screenshot of Point of Sale Part 2 for Cashier 46

20 Figure 19. Screenshot of Receipt..... 46

21 Figure 20. Screenshot of Recommendation of the System..... 47

LIST OF APPENDIX FIGURE

Appendix Figure **Page**

1 Five Brothers Trading Establishment 48

2 Five Brothers Trading Location Map 48

LIST OF APPENDICES

Appendix **Page**

1 Unit Testing 28

2 System Testing 31

3 Acceptance Testing 34

INVENTORY MANAGEMENT WITH POINT OF SALE SYSTEM WITH FEATURED-BASED RECOMMENDATION ALGORITHM FOR FIVE BROTHERS TRADING IN CARMONA, CAVITE

Lester John M. Agustin
Edmon B. Nofies
John Michael C. Paguibitan
Ashley Rovell D. Sison

An undergraduate thesis proposal submitted to the faculty of the Department of Industrial and Information Technology, Cavite State University – Carmona Campus, Carmona, Cavite in partial fulfillment of the requirements for the degree of Bachelor of Science in Computer Science. Prepared under the supervision of Mr. Jhumel C. Ignas.

INTRODUCTION

Five Brothers Trading, established in 2018, is a motorcycle parts and accessories shop located in Barangay Mabuhay, Carmona, Cavite, Philippines. The business primarily serves customers by providing various motorcycle parts, lubricants, spare components, and maintenance supplies. With an average daily transaction volume ranging from 20 to 40, the shop experiences steady demand from both regular and walk-in clients.

Despite constant business performance, Five Brothers Trading currently uses manual sales recording and inventory management procedures. Sales transactions are manually documented, generally with handwritten records or simple logs that require personnel to record each purchase separately. This process can be time-consuming and increases the chances of human errors, such as incorrect entries, lost information, or missing transaction details. Similarly, inventory management is done manually, which requires employees to physically verify stock levels and manually update records. As a result, tracking product availability, monitoring fast-moving or slow-moving products, and detecting low stock levels become inefficient and prone to

error. These traditional approaches present issues such as delayed transaction processing, inaccuracies in stock monitoring, and restricted real-time awareness of inventory levels. These challenges reduce operational efficiency and lower customer service quality.

Studies have shown that computerized Point of Sale (POS) and Inventory Management Systems can significantly enhance business operations by automating transaction documentation, providing accurate stock tracking, and improving data accessibility (Santos *et al*., 2019; Arsan, 2020). Furthermore, integrating automatic recommendation algorithms can offer tailored product suggestions, promoting upselling and better inventory turnover.

Given the nature of Five Brothers Trading's business and its existing infrastructure, the proposed system, Inventory Management with Point of Sale System using Featured – Based Recommendation Algorithm for Five Brothers Trading in Carmona, Cavite is a strategic solution. By automating daily sales and inventory processes and incorporating product recommendations, aims to reduce human errors, streamline workflows, and enhance overall business productivity.

Implementing this system at Five Brothers Trading will facilitate a seamless transition from manual to digital operations, supporting the shop's growth and improving customer satisfaction through faster and more reliable service. The system's design aligns with local business needs and infrastructure, ensuring practicality and scalability for similar small to medium-sized enterprises in the motorcycle parts industry.

This study supports Sustainable Development Goal (SDG) 8: Decent Work and Economic Growth, and SDG 9: Industry, Innovation, and Infrastructure.

Objective of the Study

The primary goal of this project is to develop and evaluate a web-based Point-of-Sale (POS) and Inventory Management System for Five Brothers Trading, following the standards outlined by the International Organization for Standardization (ISO

25010). The proposed study aims to enhance the business's ability to manage transactions, track inventory, and recommend relevant products efficiently.

Specifically, it aims to achieve the following objectives:

1. Design a system consisting of Sales and Transaction, Inventory, Recommendation, and Reports modules that support product catalog management, order processing, stock monitoring, automatic product recommendations, and auto – generated report.

2. Develop the system using Visual Studio Code with PHP, MySQL, and Python as the core technologies, supported by Figma for UI/UX design and Draw.io or Lucidchart for system modeling and diagrams.

3. Integrate the Agile Software Development Life Cycle (SDLC) as a guide for planning, designing, developing, and testing the system.

4. Implement a modern, web-based interface to streamline sales processing and improve inventory management.

5. Evaluate the system's performance using the ISO/IEC 25010 quality model in terms of product quality and quality in use.

Significance of the Study

The Inventory Management with Point of Sale System using Featured – Based Recommendation Algorithm for Five Brothers Trading in Carmona, Cavite is designed to automate and optimize the inventory and sales processes of Five Brothers Trading. Transitioning from manual operations to a digital system will significantly minimize discrepancies in stock monitoring, eliminate delays in transaction recording, and enhance the accuracy of sales and inventory data. The implementation of Featured - based Algorithm could provide insights and suggest suitable alternatives, which would enhance customer service and increasing potential sales.

Particularly, it will be beneficial to the following:

Customers. The system allows faster transactions and more accurate item availability, helping customers find the motorcycle parts they need without delays. The recommendation feature also assists them in identifying compatible or alternative products efficiently.

Staff. Employees will benefit from a user-friendly platform that reduces manual paperwork, minimizes encoding errors, and simplifies inventory updates. This efficiency will allow staff to handle customers more effectively, especially during peak hours.

Owner. The system provides real-time access to sales reports, stock levels, and recommended product insights. This enables the owner to make informed decisions regarding restocking, pricing, and inventory management. Furthermore, because the system is web-based, the owner can monitor and manage shop operations remotely through any internet-enabled device.

Future Proponents. The study will serve as a reference for future academic endeavors involving POS systems, inventory automation, and recommendation algorithms. It will contribute valuable insights into system development, data processing, and the integration of data-driven algorithms in retail environments.

Time and Place of the Study

The study will be conducted at Five Brothers Trading, located in Carmona, Cavite, where the business operations, sales transactions, and inventory processes will be observed and analyzed. Data gathering will involve interviews with the owner and staff, on-site observation of current manual procedures, and document analysis such as sales records and inventory listings.

The development, testing, and evaluation of the system will take place from September 2025 to June 2026, covering requirements analysis, system design, coding, module integration, algorithm implementation, testing, and final validation of the completed system.

Scope and Limitation of the Study

The proposed study focuses on the development, a web-based inventory management with point of sale system designed for Five Brothers Trading. The scope includes the automation of sales transactions, real-time inventory monitoring, generation of sales and stock reports, user account management, and the implementation of a Featured – Based Recommendation Algorithm that identify suggests product alternatives based on item attributes. The system will store transaction records, refresh inventory following each sale, and provide recommendations that will help staff in dealing with customers.

The system is limited to the operational requirements of a single branch and lacks advanced analytics beyond its suggestion functionality. It excludes staff management tasks such as attendance and payroll, and it does not interact with third-party platforms, including accounting software or e-commerce sites. The success of the proposal module is significantly dependent upon the accessibility and quality of sales data, thus restricting performance during the system's initial deployment. Furthermore, as the system is web-based, an internet connection is necessary to access and employ its complete capability.

Definition of Terms

Technical Terms

Database Management System (DBMS) – A software application that allows users to create, store, manage, and retrieve structured data efficiently while maintaining data integrity and security (Elmasri & Navathe, 2016).

Entity-Relationship Diagram (ERD) – A visual modeling tool used to represent the structure of a database by illustrating entities, attributes, and the relationships among them (Chen, 1976).

Feature-Based Matching – A recommendation method comparing product attributes such as type, brand, and specification.

Inventory Management System – A software system that tracks, monitors, and manages product stock levels, movements, and availability to ensure accurate inventory control and timely replenishment (Alam *et al*., 2024).

ISO 25010 – An international standard for software product quality that defines a quality model consisting of eight product quality characteristics and five quality-in-use characteristics for evaluating software systems (International Organization for Standardization, 2011).

MySQL Database – An open-source relational database management system that uses Structured Query Language (SQL) for accessing, managing, and manipulating data stored in structured tables with defined relationships (Nixon, 2021).

PHP (Hypertext Preprocessor) – A server-side scripting language designed for web development, particularly suited for database-driven web applications through its ability to process forms, access databases, and generate dynamic page content (Nixon, 2021).

Point of Sale (POS) System – A computerized system used to process sales transactions, calculate totals, generate receipts, and record sales data while automatically updating inventory levels (Saeed *et al*., 2019).

Product Alternatives – Substitute products with similar specifications used when the requested item is unavailable.

Operational Terms

Admin Operations – System activities performed by the administrator, including product management, inventory updates, user management, and report viewing.

Cashier Operations – System activities performed by the cashier, such as processing sales, viewing product details, printing receipts, and accessing recommendations.

Inventory Update - The automatic deduction or adjustment of product quantities in the system after each completed sale.

Product Recommendation – The system-generated suggestion of frequently bought-together or alternative products during a sales transaction.

Receipt Generation – The process of producing a printed or digital receipt after a successful transaction

User Authentication – The system operation that verifies the identity of an admin or cashier before granting access.

REVIEW OF RELATED LITERATURE

This chapter presents a comprehensive review of existing literature and studies that support the development of Inventory Management with Point of Sale System using Featured – Based Recommendation Algorithm for Five Brothers Trading in Carmona, Cavite. The review focuses on the concepts, methodologies, and technologies relevant to point-of-sale systems, inventory management, and recommendation algorithms.

Point of Sale Systems for Small Enterprises

Small and medium-sized firms (SMEs) are important in emerging economies, and their success frequently relies on their effective utilization of technology to enhance operations. Macatangay *(n.d.)* highlighted that efficient systems enable small enterprises to manage financial, physical, and human resources more efficiently. Manual procedures, especially in sales and inventory, are vulnerable to errors including record errors, unrecorded transactions, lost receipts, and inventory inconsistencies. These constraints affect business operations and impair the quality of customer service.

Estrellado *et al.* (2022) assert that SMEs implementing digital tools shift from reactive to proactive management. They can provide ongoing stock visibility, monitor product performance, and minimize manual redundancy. Karanja *et al.* (2024) contended that integrated digital systems improve accountability by preserving precise audit trails, guaranteeing that each transaction is recorded and traceable.

In developing countries like the Philippines, Daño-Luna *et al*. (2018) discovered that numerous SMEs continue to depend on basic instruments such as paper logs, calculators, and spreadsheets. Businesses that adopt POS systems and centralized databases exhibit enhancements in transaction speed, customer loyalty, and employee efficiency. These results demonstrate an unusual trend: small businesses utilizing POS systems significantly surpass those employing manual operations.

Point of Sale Systems

POS systems have evolved from simple cash registers into multifunctional digital platforms that automate key business processes. Tsai (2021) introduced a smart POS system for mobile retailers, showing that modern POS solutions can adapt to different business environments outside traditional storefronts. This adaptability is essential for businesses like motor shops, where transactions involve multiple product types, varying prices, and compatibility concerns. Dewi *et al*. (2021) demonstrated that using POS systems in cafés significantly improves operational efficiency by automating sales recording, inventory updates, invoice generation, and financial reporting. This reduces human error and speeds up customer transactions. Similarly, Sidhunata *et al*. (2023) designed a POS system specifically for motorcycle workshops, proving that customizing POS features to industry needs such as service tracking, parts compatibility, and stock management leads to better operational outcomes.

Additionally, POS technologies enable real-time analytics. Because transactions are stored digitally, businesses can analyze customer trends, identify best-selling products, and forecast inventory needs (Walsh & Chavan, 2020). This data-driven approach helps managers make informed decisions, improving profitability and reducing losses. Overall, the literature consistently shows that POS systems enhance accuracy, improve workflow efficiency, and provide actionable business insights.

Inventory Management System

Accurate inventory management is essential for maintaining service continuity, especially in parts-dependent businesses like motor shops. Alam *et al*. (2024) emphasized that SMEs experience significant inefficiencies when inventory is manually updated, including stockouts, overstocking, and delays in production or service delivery.

Khan *et al*. (2022) developed an IoT-enabled warehouse management system that monitors inventory in real-time by dividing warehouses into logical domains. Their

findings showed that automated systems greatly reduced stock misplacement, improved product traceability, and minimized latency in inventory retrieval.

In the Philippine context, manual stock cards and handwritten logs remain prevalent in SMEs (Tanaman *et al*., 2023). Their study on a web-based inventory system revealed that digital inventory solutions reduce discrepancies, automate stock updates, and improve the accuracy of reordering decisions. These systems especially benefit businesses handling a large variety of items, such as motorcycle parts, which differ in size, brand, compatibility, and specifications.

Inventory management literature also highlights how automated systems support loss prevention by providing accurate, real-time monitoring (Prasetyo & Gunawan, 2020). With more transparency, businesses can track missing items, identify common issues, and maintain better control over inventory.

Feature-Based Product Recommendations System

Feature-based recommendation systems operate by examining product attributes such as size, model, material, compatibility, and specifications to propose appropriate alternatives when the preferred item is not accessible. Han, Kamber, and Pei (2012) argues that content-based filtering is essential for discovering product similarities through the analysis of their attributes, thus facilitating precise substitution recommendations. Verma and Singh (2020) emphasized that feature-based models markedly enhance consumer satisfaction by providing relevant options, especially in areas where product compatibility is crucial, such as electronics, automobile components, and motorcycle parts.

Agile Development Method

Agile methodology promotes iterative development, continuous improvement, and active collaboration among developers and clients. Roque (2021) described Agile as an approach focused on incremental delivery, allowing developers to refine system features step-by-step. This fosters early identification of problems and immediate implementation of improvements.

Dewi (2021) stressed the importance of transparency in Agile, enabling stakeholders to observe progress at every stage. This is ideal for small teams working on systems that require frequent refinement, such as a motor shop POS.

Hoyos and Restrepo-Calle (2018) found Agile particularly effective for small-scale, value-oriented software projects because it reduces risk and minimizes development delays. Rapid prototyping allows users to test system modules early, ensuring the final system meets actual business needs.

Given that Five Brothers Trading requires flexibility, adaptability, and continuous refinement of its POS and recommendation systems, Agile is the most appropriate development model.

Synthesis Review of the Study

Throughout the examined literature and studies, a common theme is clear: technology significantly enhances the efficiency, accuracy, and effectiveness of company operations, particularly in small businesses. Point of Sale systems accelerate transaction processing, ensure systematic sales documentation, and facilitate dependable client assistance. Inventory management systems lessen mistakes resulting from manual tracking and guarantee that stock levels are consistently monitored and updated. Moreover, feature-based recommendation methods enhance product suggestions by comparing product attributes and identifying suitable alternative items with similar characteristics.

The literature recommends Agile development as the most effective method for small software projects requiring flexibility, fast iteration, and customer-focused refinement. These findings support the design and development of inventory system and point of sale with a feature-based recommendation engine specifically designed for motor shop operations. The solution conforms to global and local trends in digital transformation, providing data-driven insights, optimized operational workflows, and increased customer satisfaction for Five Brothers Trading.

METHODOLOGY

Material

Inventory Management with Point of Sale System will be developed using *Visual Studio Code* as the primary coding environment, with *PHP, HTML*, *CSS*, and *JavaScript* serving as the main languages for building the interface and system functionalities. *MySQL* will be used as the database management system, while *XAMPP* will function as the local server for backend configuration and testing. *Bootstrap* will be employed to ensure a responsive and user-friendly interface across all modules. *PHP* will be utilized feature-based recommendation mechanism will be implemented within the system to identify and suggest alternative products based on similar attributes such as product type, brand, or category. This approach allows the system to provide relevant product options during sales transactions while supporting inventory management. *Draw.io* and *Lucidchart* will be used to create system diagrams, including the Entity Relationship Diagram (ERD), Data Flow Diagrams (DFD), and Use Case Diagrams. Additionally, Microsoft Office applications will be used for documentation, data organization, and preparation of system reports.

The hardware that will be used includes one laptop running in Windows 11. The device is a *Lenovo IdeaPad Slim 3* equipped with an *Intel Core i5, 16GB RAM*, which will serve as the primary development. An *HP 2700* series printer will also be used to generate printed copies of documentation drafts, progress reports, and other system-related files.

Methods

The Agile method (Fig. 1) will be used as the system development approach for Inventory Management with Point of Sale System using Featured - Based Recommendation Algorithm. It is composed of six phases, namely: Requirements, Design, Development, Testing, Deployment, and Review. Agile is known for delivering working software in regular intervals, prioritizing the most valuable features first (R. V.,

A., & Dinakaran, n.d.). This method can be tailored to meet the specific needs of Five Brothers Trading by limiting work in progress, maintaining flexibility, and focusing on continuous improvement. Through this approach, the development team can respond quickly to changes, refine system features based on client input, and ensure that each iteration directly addresses identified business challenges. The adoption of Agile allows to progress smoothly through the development cycle, ensuring efficient updates, transparency between developers and the motor shop staff, and consistent integration of user feedback. These practices significantly support the creation of a reliable and user-centered inventory management and point of sale system for Five Brothers Trading.

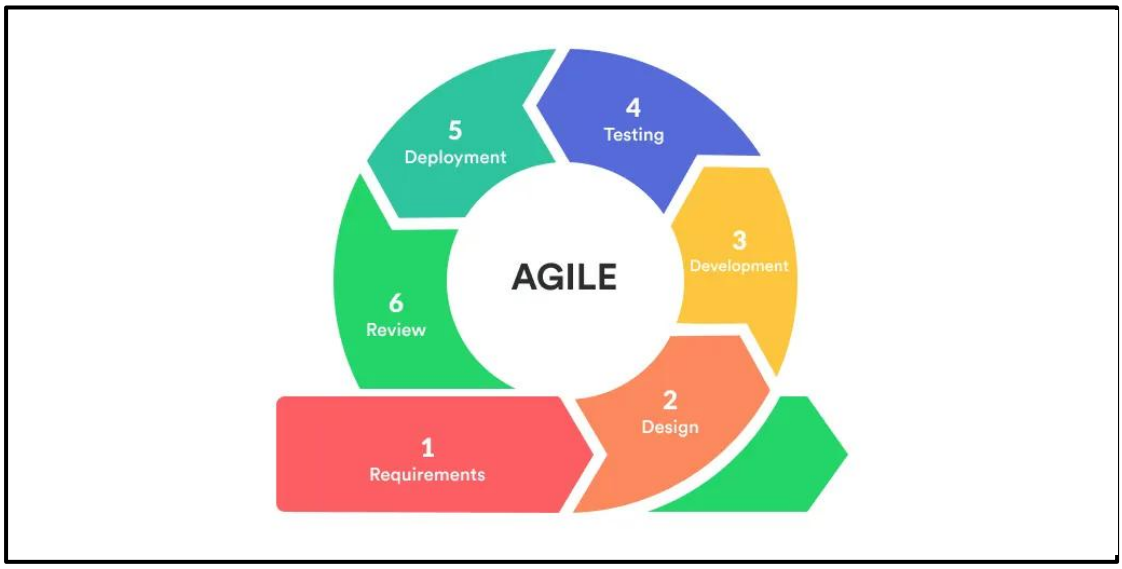


Figure 1. Agile Model (Eby, 2017)

The Agile process begins with requirements, where the proponents identify the problems within the current manual workflow of Five Brothers Trading, including issues in stock checking, sales recording, and the absence of product recommendations. The design phase follows, involving the creation of interface layouts and system flow structures that emphasize usability and clarity for motor shop staff. The development phase focuses on building the system's key functionalities, such as POS operations, real-time inventory management, and the Featured - Based Recommendation Algorithm. Testing is performed throughout development to ensure accuracy in sales

processing, reliability in inventory updates, and consistency in generating recommendations. The deployment phase includes launching the system for actual use at Five Brothers Trading, followed by monitoring and maintenance to ensure stable performance and continuous improvement. Finally, the review phase involves gathering insights from the owner and staff to refine system features and address any issues

Requirements. The requirements analysis phase began with the proponents visiting the beneficiary, Five Brothers Trading, to observe its current workflow and identify the existing issues associated with its manual sales and inventory processes. During the visit, the proponents gathered detailed information regarding daily transactions, stock monitoring routines, and the challenges encountered by the staff in handling customer requests (Appendix 1). This phase aimed to establish a clear understanding of the project scope, determine the system requirements, and outline the features needed to address the problems found in the motor shop's operations.

The manual process relies heavily on handwritten documents, memory-based price checking, and physical inspection of stocks, which often leads to delays, inconsistencies, and inaccuracies. After documenting the workflow, the proponents prepared an Input-Process-Output (IPO) diagram (Fig. 2) to illustrate the current system and highlight areas that require improvement.

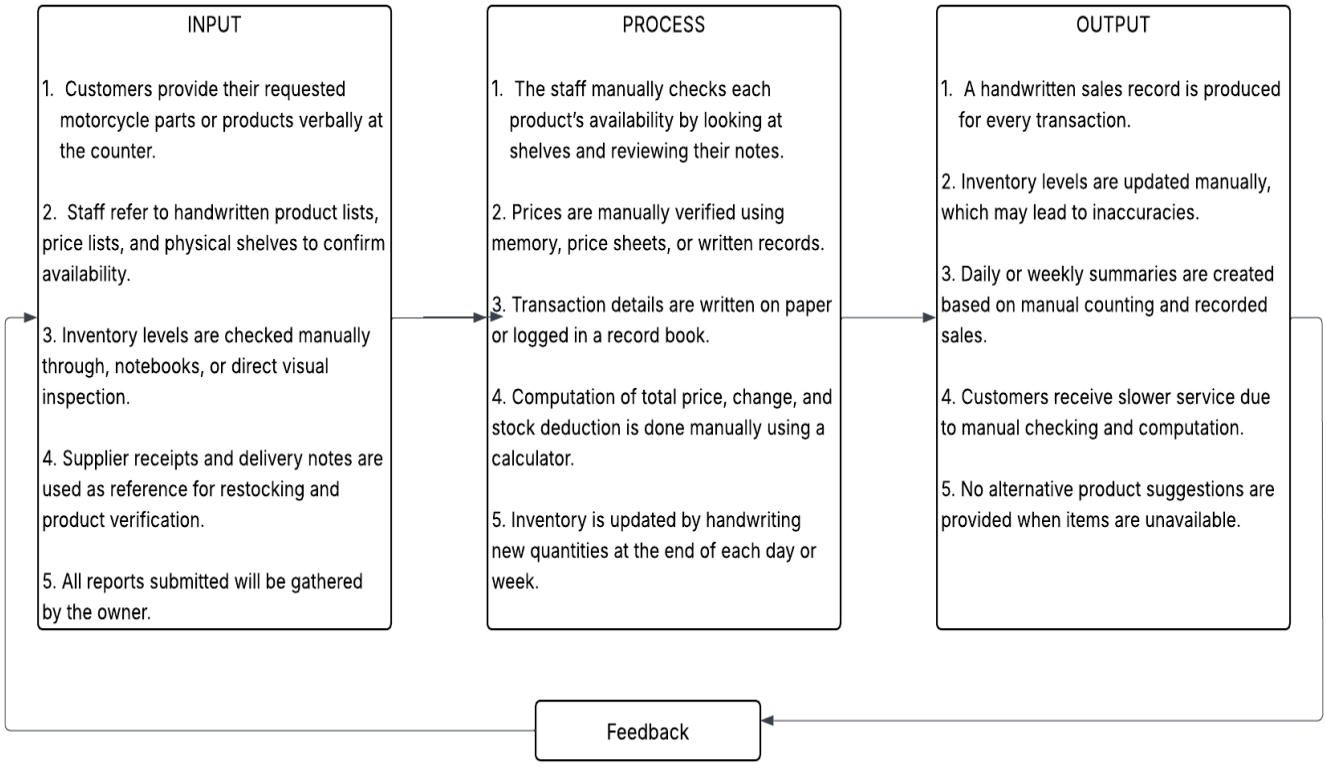


Figure 2. Input-Process-Output (IPO) diagram of the manual system

The IPO diagram of the manual process (Fig. 2) illustrates how Five Brothers Trading currently manages its sales and inventory operations using handwritten documents and manual checking. Product availability, pricing, and stock levels are verified through physical inspection and notes, while computations and record keeping are performed manually using calculators and logbooks. This workflow is prone to human error, delays, and inconsistent inventory updates, resulting in slower service and unreliable reports. The absence of automated recommendations also limits the staff's ability to assist customers when a product is unavailable. These issues highlight the need for a computerized system capable of improving accuracy, efficiency, and service quality.

Design. During the design phase, the proponents transformed the identified requirements into system models and structured designs. An IPO diagram of the proposed system (Fig. 3) was developed to illustrate how inputs such as customer purchases, product data, and sales history are processed automatically to generate

accurate outputs. This diagram demonstrates how the system integrates POS operations, real-time inventory management, and the Featured-Based Recommendation Algorithm.

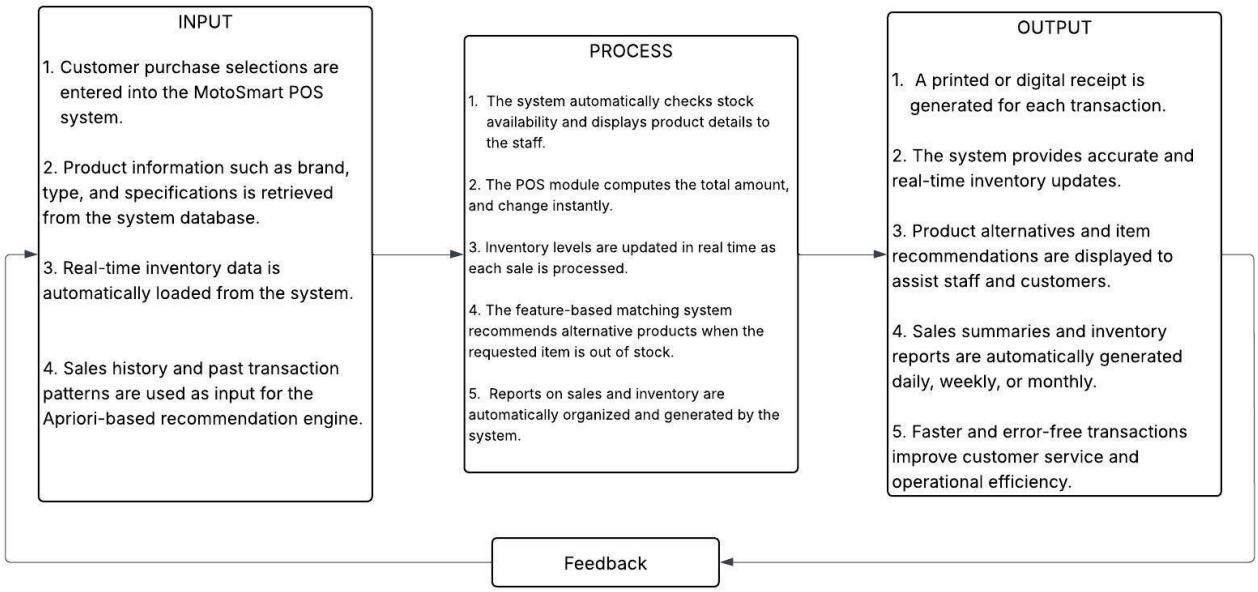


Figure 3. IPO diagram of the Inventory Management with Point of sale System using Featured-Based Recommendation Algorithm for Five Brothers Trading in Carmona, Cavite

A Use Case Diagram (Fig.4) was created for Inventory Management with Point of sale System using Featured-Based Recommendation Algorithm for Five Brothers Trading in Carmona, Cavite. This diagram outlines the system's high-level functions and identifies which modules can be accessed by each actor, specifically the Admin and the Cashier. The design clarifies how users will interact with the system and establishes the foundation for building the core functionalities.

The client's involvement remains active throughout this phase, allowing them to review the initial design and suggest revisions as necessary. As the Agile methodology emphasizes iterative improvement, the proponents continuously refine the system design based on the client's feedback. The primary focus of the early design cycles is to establish the essential POS and inventory features, such as product management, stock monitoring, sales processing, and automated recommendations.

Once the diagram and interface concepts are drafted, the proponents develop a prototype and present it to the client for evaluation. Feedback is gathered and applied to enhance usability, accuracy, and workflow alignment. This iterative process is repeated until the design fully meets the operational needs of Five Brothers Trading.

Use Case Diagrams in Unified Modeling Language (UML) provide a visual representation of system behavior, showing how actors interact with various functions of the application. While the diagram does not show internal logic or technical implementation, it effectively communicates the system’s scope, user roles, and required interactions, serving as a crucial guide for both development and client validation. The following diagram shows the system's three user types: administrator, cashier, and staff. The admin has limited administrative access and is primarily responsible for generating and managing user accounts, as well as observing general sales and inventory records for monitoring purposes. The Cashier handles operational activities such processing sales transactions, viewing product details, producing receipts, monitoring stock information, and accessing system-generated product recommendations during transactions. Meanwhile, the Staff is in charge of inventory management tasks such as adding, updating, modifying, and removing product records to ensure inventory data is accurate and up to date. This clear separation of responsibilities contributes to an organized, secure, and efficient workflow at Five Brothers Trading.

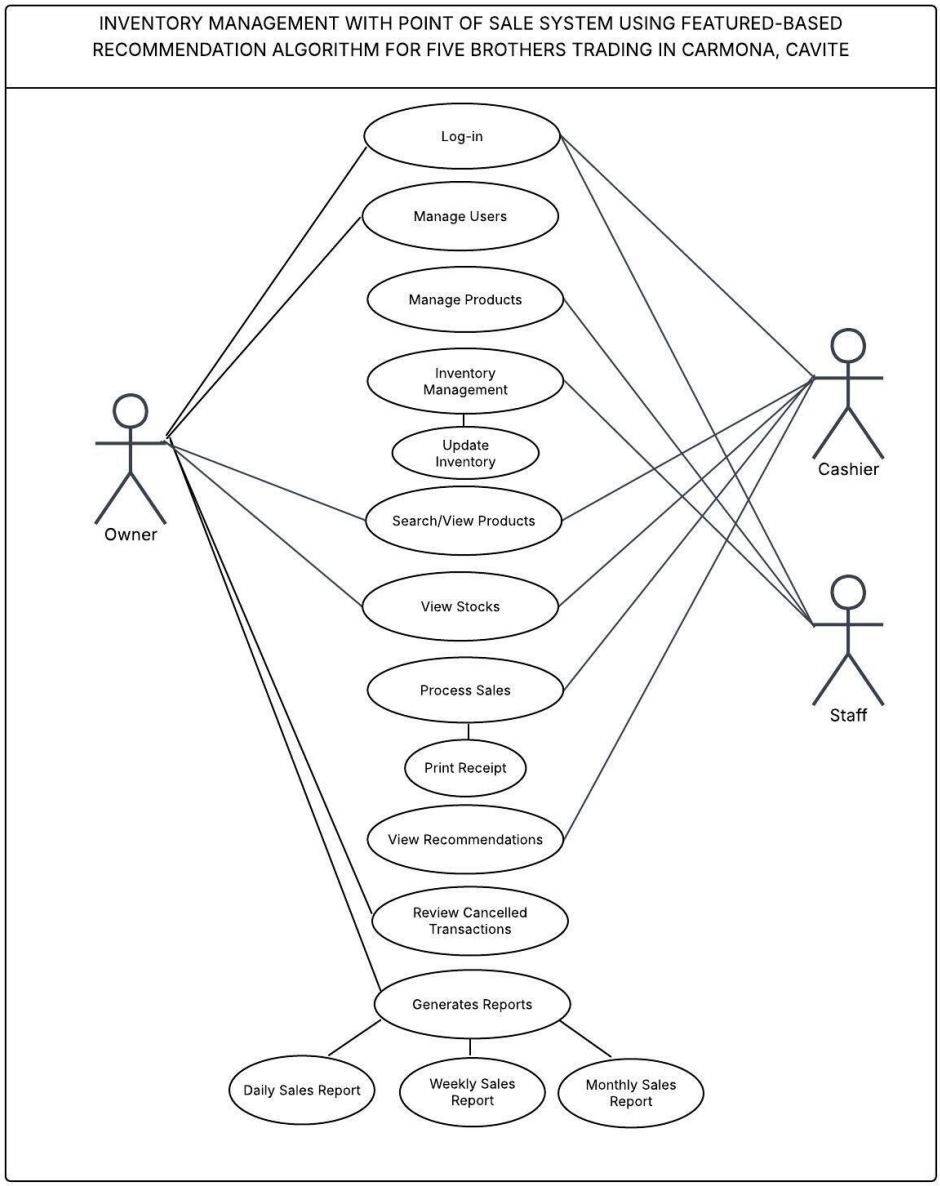


Figure 4. Use Case Diagram of the Development of Inventory Management with Point of sale System using Featured-Based Recommendation Algorithm for Five Brothers Trading in Carmona, Cavite

The ERD (fig.5) illustrates the complete data structure of the proposed system, showing how entities interact to support sales processing, inventory monitoring, user management, and the recommendation algorithm. It serves as the blueprint for database design, ensuring that all modules of the system are integrated, organized, and capable of handling real-time operations.

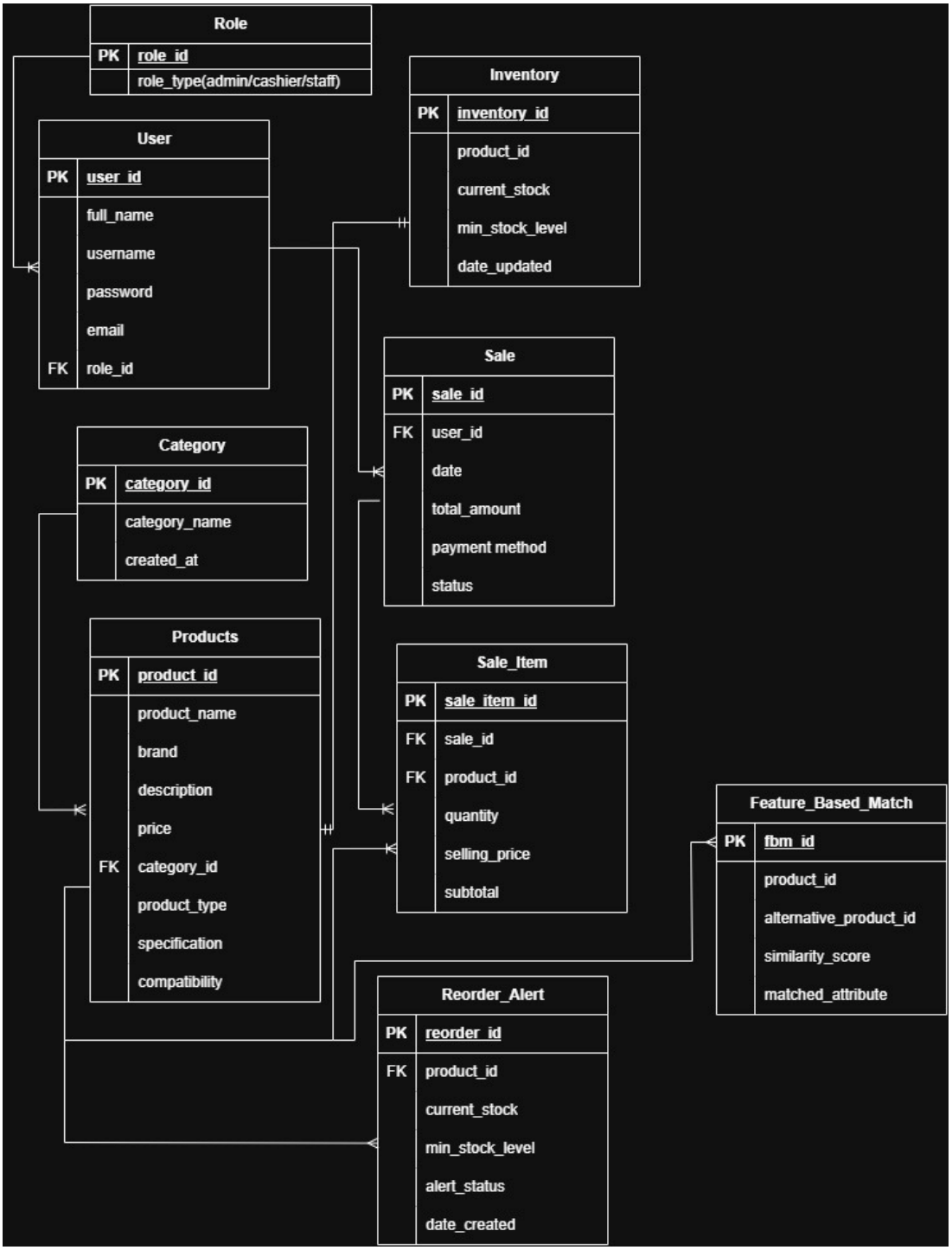


Figure 5. Entity Relationship Diagram (ERD) of the Inventory Management with Point of sale System using Featured-Based Recommendation Algorithm for Five Brothers Trading in Carmona, Cavite

A Feature-Based Recommendation Algorithm Diagram (fig. 6) was developed for the *Inventory Management with Point of Sale System for Five Brothers Trading in Carmona, Cavite*. This diagram shows how the system generates product recommendations during a sales transaction. When a product is selected, the system retrieves its attributes such as category and brand.

The system then performs feature-based matching to identify similar products from the database. A filtering process is applied to exclude out-of-stock and identical items, ensuring that only available and relevant products are suggested.

The remaining products are then organized and displayed in the Point of Sale interface as alternative options. This approach provides simple and effective recommendations while supporting inventory management and improving transaction efficiency.

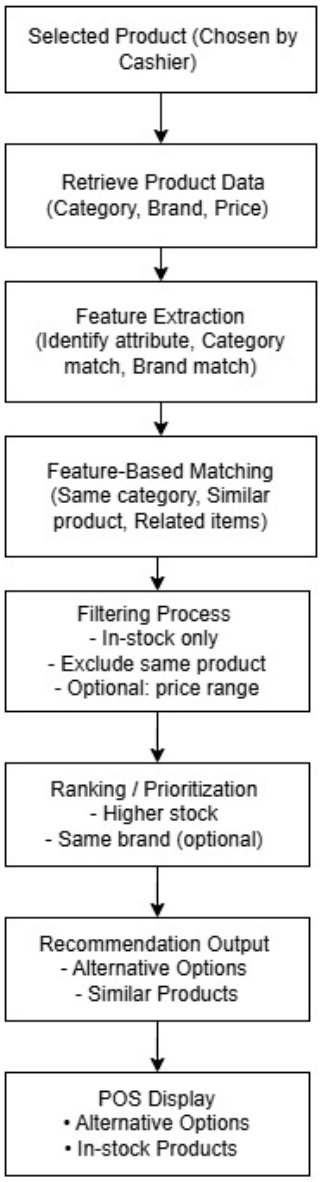


Figure 6. Algorithm Diagram of the Inventory Management with Point of sale System using Featured-Based Recommendation Algorithm for Five Brothers Trading in Carmona, Cavite

Development. The development phase focused on implementing the system based on the approved design models. Core modules such as user authentication, sales processing, product management, and real-time inventory updates were developed first. The Featured-Based Recommendation Algorithm was then integrated to analyze historical transaction data and generate alternative product suggestions when items are unavailable.

Development will be conducted in iterative cycles, allowing continuous refinement of system features based on testing results and stakeholder feedback.

Testing. This will ensure that the software met all the requirements and performed as expected. Testing was implemented throughout the development process to ensure that the product was working as expected with optimal quality. The proponents will also gather feedback from the Five Brothers owner to distribute the system and questionnaire to the respondent.

Unit testing. The Unit Testing (Appendix 2) will ensure that the system objectives will be met. This testing focuses on ensuring that the modules and features are working properly.

System testing. An Evaluation Form (Appendix 3) adopted from the International Organization for Standardization (ISO) 25010 will be used to ensure the system meets the expected quality in terms of functional stability, performance efficiency, compatibility, compatibility, usability, reliability, security, maintainability, and portability. Within this context, functional suitability (indicators 1 to 3) evaluates whether the system provides the required features to support Five Brothers Trading operations. This includes the availability of core modules such as inventory, sales and reports, and settings. The system is assessed based on its ability to process sales transactions accurately, manage product and inventory records efficiently, generate reports automatically, and produce receipts for each transaction.

The performance efficiency criterion (indicators 4 to 6) measures the system's responsiveness when performing tasks such as product selection, logging in and out, page navigation, and executing action buttons. It also evaluates the speed and accuracy of CRUD operations, real-time inventory updates, and response time during sales transactions.

Compatibility (indicators 7 to 8) testing checks the system's ability to run properly across different web browsers and operate consistently on various devices without performance issues.

Usability criteria (indicators 9 to 19) assess the system's ease of use, clarity of interface, and consistency in design. This includes evaluating navigation labels, ease of learning for new users, error prevention, accessibility of information, and overall visual appearance.

Reliability (indicators 20 to 22) focuses on the system's ability to perform transactions accurately, maintain stable operation during extended use, and recover smoothly from minor errors.

Security criteria (indicators 23 to 27) evaluate the system's protection of data through controlled access, prevention of unauthorized modifications, secure storage of user credentials, and protection of transaction data

Maintainability (indicators 28 to 30) measures how easily the system can be tested and how reusable its components are for future system improvements or other applications.

Portability (indicators 31 to 33) assesses the system's ability to run on different web browsers such as Google Chrome and Microsoft Edge, its ease of installation, and its capability to replace similar software in the same environment.

The respondents were chosen using the convenience sampling

technique. This non-probability sampling method involves choosing participants based on their convenient availability and willingness to participate in the study (Nikolopoulou, 2023).

A 5-point Likert scale will be used in the evaluation: 1-poor, 2-fair, 3-satisfactory, 4-very satisfactory, and 5- Outstanding. The mean of the

responses will be used to represent the typical value in the dataset. It will use the formula below.

$$\bar{x} = \frac{\Sigma(wx)}{\Sigma w}$$

where:

$\bar{x}$ = mean

Σ = summation

w = the weights

x = the value

Table 1. Mean Interpretation Table

VALUE	LIMITS	DESCRIPTIVE EQUIVALENT
5	4.50 - 5.00	Outstanding
4	3.50 – 4.49	Very Satisfactory
3	2.50 – 3.49	Satisfactory
2	1.50 – 2.49	Fair
1	1.00 – 1.49	Poor

The interpreted values of the computed weighted mean for each indicator were based on the mean interpretation table established by Mangubat, Mangahas, Matias, and Mauden in 2014.

Standard Deviation will also be used and computed to measure the dispersion of the data from the mean. The formula for standard deviation is:

$$SD = \sqrt{\frac{\Sigma(x - \mu)^2}{N}}$$

where:

SD = standard deviation

Σ = summation

x = value in a data set

μ = mean in the data set

N = number of data points in the population

Deployment. The deployment phase involves the completed system to Five Brothers Trading. This includes installing the system, preparing the database, and configuring all modules for actual use. The proponents will conduct a demonstration for the client and train the staff on system usage. After deployment, initial monitoring will be performed to ensure system stability and address any modifications needed during the early stages of implementation.

Review. Regular feedback sessions will be conducted with the client to review updated versions of the system. This ensures that the evolving features align with the motor shop's operational needs. Adjustments in the interface, workflow, or functionalities will be incorporated based on client observations. Agile encourages frequent meetings and open communication, allowing the proponents to refine the system continuously based on real-world expectations.

REFERENCES

Agrawal, R., & Srikant, R. (1994). Fast algorithms for mining association rules. Proceedings of the 20th International Conference on Very Large Data Bases (VLDB), 487–499.

Agustin, L. *(n.d.).* Point of sale system with inventory for Arm's Food and Delicacies [PDF file]. Retrieved from https://www.researchgate.net/profile/Louie-Agustin/publication/337707746_POINT_OF_SALE_SYSTEM_WITH_INVENTORY_FOR_ARM%27S_FOOD_AND_DELICACIES/links/5de653e2a6fdcc2837010d26/POINT-OF-SALE-SYSTEM-WITH-INVENTORY-FOR-ARMS-FOOD-AND-DELICACIES.pdf

Choubey, D. *(n.d.).* Implementation of inventory management system [PDF file]. Retrieved from https://www.researchgate.net/profile/Dilip-Choubey-2/publication/341075161_Implementation_of_Inventory_Management_System/links/5eac148692851cb267693e6e/Implementation-of-Inventory-Management-System.pdf

Choudhury, T. M. (2022). Web-based POS and inventory management system for SMEs, retail stores, and restaurants (Internship Report). Independent University, Bangladesh. Retrieved from http://103.81.70.85/handle/11348/787

Dalal Attal, M. N., *et al*. (2018). Redesigning a retail store based on association rule mining. Proceedings of the *International IEOM Conference Paris* 2018, 370–375.

Goh, C. Y. (2022). Digital transformation for SMEs in emerging economies. Journal of Business and Innovation, 9(2), 55–68.

Google Research. (2020). Teachable Machine: Train a computer to recognize your own images, sounds, & poses. Retrieved from https://teachablemachine.withgoogle.com/

Goodfellow, I., Bengio, Y., & Courville, A. (2016). Deep learning. MIT Press.

Han, J., Kamber, M., & Pei, J. (2012). Data mining: Concepts and techniques (3rd ed.). Morgan Kaufmann.

ISO. (2011). ISO/IEC 25010: Systems and software engineering – Systems and software Quality Requirements and Evaluation (SQuaRE). International Organization for Standardization.

Jayathilaka, C. (2020, July 30). Agile methodology. Medium. https://medium.com/@chathmini96/agile-methodology-30ec4cdf3f

Macatangay, A. A. *(n.d.).* Review of Related Literature. Scribd. Retrieved November 6, 2025, from https://www.scribd.com/document/635721932/Untitled

Madamidola, O. *(n.d.).* A review of existing inventory management systems [PDF file]. Retrieved from https://www.researchgate.net/profile/Olugbenga-Madamidola/publication/383947700_A_Review_of_Existing_Inventory_Mana

gement_Systems/links/66e34077f84dd1716ce9ccee/A-Review-of-Existing-Inventory-Management-Systems.pdf

Magno, A. E. F., Belo, P. A., & Miranda, C. M. (2022). An Agile approach to a project management information system amidst the COVID-19 pandemic. Innovatus: Digital Transformation in Business Information Systems, 5(1). https://innovatus-pub.github.io/abstractpublications_archive/2021/paper2_pdf.pdf

Mariani, M. J. P., & Wacas, R. U. (2022). Efficacy, reliability, and accuracy of inventory system and point-of-sale service for small-medium enterprises in Tabuk City. International Journal of English Literature and Social Sciences, 7(6), 306–308. https://doi.org/10.22161/ijels.76.43

Mensouri, D. (2022). A new marketing recommendation system using a hybrid model. Acta Computer Science and Systems, 2(1), 16–25.

Praditya, N. W. P. Y. (2021). Literature review: Recommendation system using hybrid models. CORE Repository.

Roni, M. S., Jin, M., & Eksioglu, S. D. (2015). A hybrid inventory management system responding to regular demand and surge demand. Omega, 52, 190–200. https://doi.org/10.1016/j.omega.2014.05.002

Suresh, A. *(n.d.).* A comprehensive study of hybrid recommendation systems for e-commerce applications [PDF file]. Retrieved from https://www.researchgate.net/profile/A-Suresh-3/publication/349585743_A_Comprehensive_Study_of_Hybrid_Recommendation_Systems_for_E-Commerce_Applications/links/6037326392851c4ed59543ea/A-Comprehensive-Study-of-Hybrid-Recommendation-Systems-for-E-Commerce-Applications.pdf

Wanaskar, U. R., Vij, S., & Mukhopadhyay, D. (2013). A hybrid web recommendation system based on the clustering of frequent itemsets. *International Journal of Computer Applications*, 84(9), 28–33.

Wanaskar, U., Vij, S., & Mukhopadhyay, D. (2013). A hybrid web recommendation system based on the improved association rule mining algorithm. *International Journal of Computer Science and Information Technologies*, 4(3), 423–428.

APPENDICES

Appendix 1

Interview Guide

Greetings! As part of proposed system, we, a group of 4th-year college students pursuing the Bachelor of Science on Computer Science program, are excited to collaborate with you. Your valuable insights are crucial for the development of an efficient MotoSmart: Point of Sale and Inventory Management System with Hybrid Apriori-Based Recommendation Algorithm for Five Brothers Trading in Carmona, Cavite.

Rest assured, all information shared during the interview will be treated with the utmost confidentiality, adhering to the Data Privacy Law of 2012. The collected data will be used exclusively for the purpose of refining and developing the POS and Intentory system.

Respondent's Name: Fervan Barabat

Date: Nov 20, 2025

Venue: Five Brothers Trading

1. Can you provide a brief overview of the business, including the products you sell and your daily operations?	We sell motorcycle parts and accessories. Everyday we assist walk in customers and manage stock deliveries.
2. What are the key challenges you currently experience in managing sales and inventory?	It's hard to keep track of all sales and know exactly what stock is left without a system.
3. What is the average volume of orders on a daily, weekly, and monthly basis?	We get around 20 orders daily, about 100 weekly and roughly 400 monthly.
4. How do you currently track and manage your inventory, and what difficulties do you encounter using this method?	We manually write down inventory counts, which can lead to errors and delays.
5. How do you monitor stock levels and identify products that are low in quantity or out of stock?	We check stock physically and sometimes find out late when items are already out.
6. How are sales records, receipts, and reports currently prepared and stored?	Sales are recorded in notebooks, receipts are handwritten and reports are made monthly but take time
7. How do you handle situations when customers request products that are unavailable, and are alternatives suggested?	We inform customers about stockouts and sometimes suggest similar products if available.

Appendix 2

Unit Testing

Republic of the Philippines
CAVITE STATE UNIVERSITY
CvSU Carmona Campus
Carmona, Cavite
(046) 487-6328
www.cvsu.edu.ph

EVALUATION INSTRUMENT FOR INVENTORY MANAGEMENT WITH POINT OF SALE SYSTEM USING FEATURED-BASED RECOMMENDATION ALGORITHM FOR FIVE BROTHERS TRADING IN CARMONA, CAVITE

Dear Participant,

Good day! We currently conducting research entitled "**INVENTORY MANAGEMENT WITH POINT OF SALE SYSTEM USING FEATURED-BASED RECOMMENDATION ALGORITHM FOR FIVE BROTHERS TRADING IN CARMONA, CAVITE**". In line with this, we are respectfully requesting your assistance in filling out this evaluation form. It will not be a problem if you wish not to participate, but your responses will be highly valued. The evaluation form can be completed anonymously. Responses from completed questionnaires will be collated for analysis; once complete, the original questionnaires will be kept electronically. Rest assured that all information indicated therein will be treated with utmost confidentiality under the Data Privacy Law of 2012 and strictly used only for the above purpose. All the gathered information/data will also be retained in the system and used as a part of the historical data for further analysis. If you wish to learn more about the results of the research, please send an email to **cc.ashleyrovell.sison@cvsu.edu.ph**

We are hoping for your kind consideration and support. Thank you very much.

Name (Optional): _____ Date: _____
Address (Optional): _____
Profession: _____ Specialization: _____

Instructions: Please evaluate using the given scale and placing a checkmark (✓) on the appropriate column corresponding to your response.

Numerical Rating:
5 – Excellent 4 – Very Good 3 – Good 2 – Fair 1 – Poor

INDICATOR	5	4	3	2	1
Effectiveness					

INDICATOR	5	4	3	2	1
1. **Effectiveness** - The system allows users to achieve specified goals with accuracy and completeness.					
Efficiency					
2. **Efficiency** - The system requires minimal resources expended in relation to the accuracy and completeness of goal achievement.					
Satisfaction					
3. **Usefulness** - The user is satisfied with their perceived achievement of pragmatic goals, including the results of use and the consequences of use.					
4. **Trust** - The user or other users has confidence that the system will behave as intended.					
5. **Pleasure** - The user obtains pleasure from fulfilling their personal needs through the system.					
6. **Comfort** - The user is satisfied with physical comfort while using the system.					
Freedom from Risk					
7. **Economic Risk Mitigation**- The system mitigates the potential risk to financial status, efficient operation, commercial property, reputation or other resources.					
8. **Health and Safety Risk Mitigation**- The system mitigates the potential risk to people (health and safety) in the intended contexts of use.					
9. **Environmental Risk Mitigation**- The system mitigates the potential risk to property or the environment in the intended contexts of use.					
Context coverage					
10. **Context Completeness**- The system can be used with effectiveness, efficiency, freedom from risk, and satisfaction in all the specified contexts of use.					
11. **Flexibility**- The system can be used with effectiveness, efficiency, freedom from risk and satisfaction in contexts beyond those initially specified.					

Adopted from the International Organization for Standardization (ISO) 25010 for quality in use

Remarks/Comments/Suggestions:

_____

_____

_____

_____

_____
Signature of Respondent

Appendix 3

System Testing

Republic of the Philippines
CAVITE STATE UNIVERSITY
CvSU Carmona Campus
Carmona, Cavite
(046) 487-6328
www.cvsu.edu.ph

EVALUATION INSTRUMENT FOR
INVENTORY MANAGEMENT WITH POINT OF SALE SYSTEM USING FEATURED-BASED RECOMMENDATION ALGORITHM FOR FIVE BROTHERS TRADING IN CARMONA, CAVITE

Dear Participant,

Good day! We currently conducting research entitled "**INVENTORY MANAGEMENT WITH POINT OF SALE SYSTEM USING FEATURED-BASED RECOMMENDATION ALGORITHM FOR FIVE BROTHERS TRADING IN CARMONA, CAVITE**". In line with this, we are respectfully requesting your assistance in filling out this evaluation form. It will not be a problem if you wish not to participate, but your responses will be highly valued. The evaluation form can be completed anonymously. Responses from completed questionnaires will be collated for analysis; once complete, the original questionnaires will be kept electronically. Rest assured that all information indicated therein will be treated with utmost confidentiality under the Data Privacy Law of 2012 and strictly used only for the above purpose. All the gathered information/data will also be retained in the system and used as a part of the historical data for further analysis. If you wish to learn more about the results of the research, please send an email to **cc.ashleyrovell.sison@cvsu.edu.ph.**

We are hoping for your kind consideration and support. Thank you very much.

Name (Optional): _____ Date: _____
Address (Optional): _____
Profession: _____ Specialization: _____

☐ IT expert ☐ Management
☐ Customer ☐ Cashier

Instructions: Please evaluate using the given scale and placing a checkmark (✓) on the appropriate column corresponding to your response.

Numerical Rating:
5 – Excellent 4 – Very Good 3 – Good 2 – Fair 1 – Poor

INDICATOR	5	4	3	2	1
Functional Suitability					
1. **The system has the following modules:** **a. inventory,** **b. sales and report** **c. settings**					
2. **The system can be used to:** **a. Process sales transactions accurately.** **b. Manage product and inventory records efficiently.** **c. Generate sales and inventory reports automatically.**					
3. **The system can generate a receipt.**					
Performance Efficiency					
4. **The system provides an appropriate response time when performing various tasks such as:** **a. Selecting products from the inventory** **b. Logging in and out** **c. Clicking navigation buttons** **d. Clicking actions buttons to perform specific actions** **e. CRUD: view, add, edit, modify, delete, search, and filter system contents.**					
5. **The system efficiently updates inventory records in real time.**					
6. **The system responds within an acceptable time during sales transactions.**					
Compatibility					
7. **The system runs properly on different web browsers.**					
8. **The system operates consistently across devices.**					
Usability					
9. **The system interface is easy to understand.**					
10. **The navigation buttons are clearly labeled.**					
11. **The system is easy to learn for new users.**					
12. **The system minimizes user errors.**					
13. **The system has a consistent and user-friendly interface.**					
14. **The required information is easy to locate.**					
15. **The interface looks good.**					
Reliability					
16. **The system performs transactions accurately without data loss.**					
17. **The system maintains consistent operation during extended usage.**					
18. **The system recovers smoothly from minor errors.**					
Security					

INDICATOR	5	4	3	2	1
19. **The system ensures that data are accessible only to those authorized to have access.**					
20. **The system prevents unauthorized access to, or modification of, computer programs or data.**					
21. **User credentials are securely stored and encrypted.**					
22. **The system protects transaction data.**					
Maintainability					
23. **The system can easily test.**					
24. **The asset/system can be used in more than one system or for building other assets.**					
Portability					
25. **The system can be moved to different environments such as:** **a. Google Chrome,** **b. Microsoft Edge.**					
26. **The system can be installed easily.**					
27. **The system can replace another specified software product for the same purpose in the same environment.**					

Adopted from the International Organization for Standardization (ISO) 25010 for product quality

Remarks/Comments/Suggestions:

_____

_____

_____

_____

_____

Signature of Respondent

Appendix 4

Acceptance Testing

Republic of the Philippines
CAVITE STATE UNIVERSITY
CvSU Carmona Campus
Carmona, Cavite
(046) 487-6328
www.cvsu.edu.ph

INVENTORY MANAGEMENT WITH POINT OF SALE SYSTEM USING FEATURED-BASED RECOMMENDATION ALGORITHM FOR FIVE BROTHERS TRADING IN CARMONA, CAVITE

CRITERION	DESCRIPTION	TESTING	REMARK
Module	**ROLE SELECTION (Pre-Login)** The module will present users with a choice to select their role (Cashier, Staff or Admin) before being redirected to the appropriate login screen.	**First Test** The module will be tested by clicking the 'Cashier' role option and verifying that the system successfully navigates the user to the Cashier Login interface.	
	LOGIN (Admin) The LOGIN module will allow Admin users to securely access the system by validating their unique credentials (username and password) against the database.	**First Test** The module will be tested by entering a correct username and password and verifying that the system successfully navigates to the Admin Dashboard screen.	
	DASHBOARD (Admin) The module will display an overview of daily sales and key operational metrics for the administrator.	**First Test** The module will be tested by simulating a full sales transaction and verifying that the sales figures on the dashboard update	

CRITERION	DESCRIPTION	TESTING	REMARK
		immediately and accurately.	
	CATEGORY (Admin) The CATEGORY module will allow the admin user to define, view, edit, and manage the product categories used to organize inventory.	**First Test** The test will verify the creation of a new category by successfully entering the required details, saving the entry, and confirming its appearance in the main table.	
	PRODUCTS (Admin) The PRODUCTS module will enable the admin user to manage the entire product inventory, including adding new stock, updating pricing, and viewing stock levels.	**First Test** The module will be tested in the product management functionality by adding a new product entry, including its category, price, and initial stock quantity, and verifying all data fields are correctly saved.	
	USERS (Admin) The USERS module will provide the administrative interface for managing all system accounts, allowing the admin to view, create, and modify user credentials and access roles.	**First Test** The test will attempt to create a new Cashier user account, verify that the account is stored, and successfully modify the user's role or status.	
	RECORDS (Admin) The RECORDS module will enable the admin user to	**First Test** The test will verify that the module can successfully	

CRITERION	DESCRIPTION	TESTING	REMARK
	view and generate pre-defined financial and inventory reports based on transactional data for analysis and record-keeping.	generate a report (a sales report) for a specified date range and that the resulting data is correctly collected and displayed.	
	LOGIN (Cashier) The LOGIN module will allow Cashier users to securely access the point-of-sale system by validating their unique credentials (username and password).	**First Test** The test will verify successful login by entering a valid set of Cashier credentials and checking that the system redirects to the main Cashier Dashboard.	
	DASHBOARD (Cashier) The DASHBOARD module will provide a summary of the Cashier's relevant daily statistics, such as total sales processed.	**First Test** The module will confirm that after processing several sales, the dashboard correctly updates the cashier's daily sales count and total revenue for the current shift.	
	CATEGORY (Cashier) The CATEGORY module will allow Cashier users to **view** the list of product categories only, with no creation, editing, or deletion functionality.	**First Test** The module will be tested by a logged-in Cashier accessing the Category screen and verifying that the list of categories is displayed, but that no 'Add', 'Edit', or 'Delete' controls are visible or functional.	

CRITERION	DESCRIPTION	TESTING	REMARK
	PRODUCTS (Cashier) The PRODUCTS module will enable the Cashier user to **view** the product inventory, including stock levels and pricing, to support the sales process, with no administrative functions available.	**First Test** The module will be tested by a logged-in Cashier accessing the Products screen and verifying that the full list of products and their details are displayed, but that no inventory modification functions are accessible.	
	TRANSACTION/POS (Cashier) The TRANSACTION module will allow the Cashier user to quickly process new sales by selecting items and calculating the bill.	**First Test** The user will test the item selection process by successfully adding several products to the transaction list and verifying the subtotal calculation is accurate.	
	PAYMENT/SALES (Cashier) The PAYMENT module will allow the Cashier to finalize a sale, including processing payment and accurately calculating any change due to the customer.	**First Test** The module will simulate payment by entering the tendered amount and verifying that the system correctly calculates and displays the change amount and finalizes the transaction.	

CRITERION	DESCRIPTION	TESTING	REMARK
	HISTORY (Cashier) The HISTORY module will allow the Cashier user to view a list of recent transactions processed through the system.	**First Test** The user will verify that the module correctly displays a list of the last ten transactions and that clicking a specific transaction loads the complete sales details.	
Security	**Database** The system will use a database (e.g., MySQL) to handle backend operations, ensuring real-time data accuracy for transaction processing and inventory storage.	**First Test** The database will be tested by performing CRUD (Create, Read, Update, Delete) operations on test records to verify that data is stored accurately and that the system maintains integrity and consistency between sales and inventory records.	
	User Access Levels The system will provide secure access levels for Admin and Staff users to ensure data confidentiality and integrity.	**First Test** The system will be tested by logging in using both Admin and Staff credentials to verify that each user type is correctly restricted to their defined functionalities and secure access levels are maintained.	
Hardware	**Standard Computing Platform**	**First Test** The system will be tested by running	

CRITERION	DESCRIPTION	TESTING	REMARK
	The system will be designed to run smoothly on standard desktop computers and laptops and will not require mobile or iOS deployment in this version.	the web application on standard desktop computers and laptops using updated web browsers (Google Chrome) to ensure the interface is responsive and runs smoothly without lag during a simulated transaction.	
Platform	**Google Chrome Browser** The web-based system will function properly on Google Chrome Browser, including all the features in the website.	**First Test** The system will be tested by logging in as an Admin and completing a full product management cycle (create, edit, delete) to verify proper rendering and functionality in the Google Chrome browser.	

Prepared by:
LESTER JOHN AGUSTIN
EDMON B. NOFIES
JOHN MICHAEL C. PAGUIBITAN
ASHLEY ROVELL D. SISON

LIST OF FIGURES

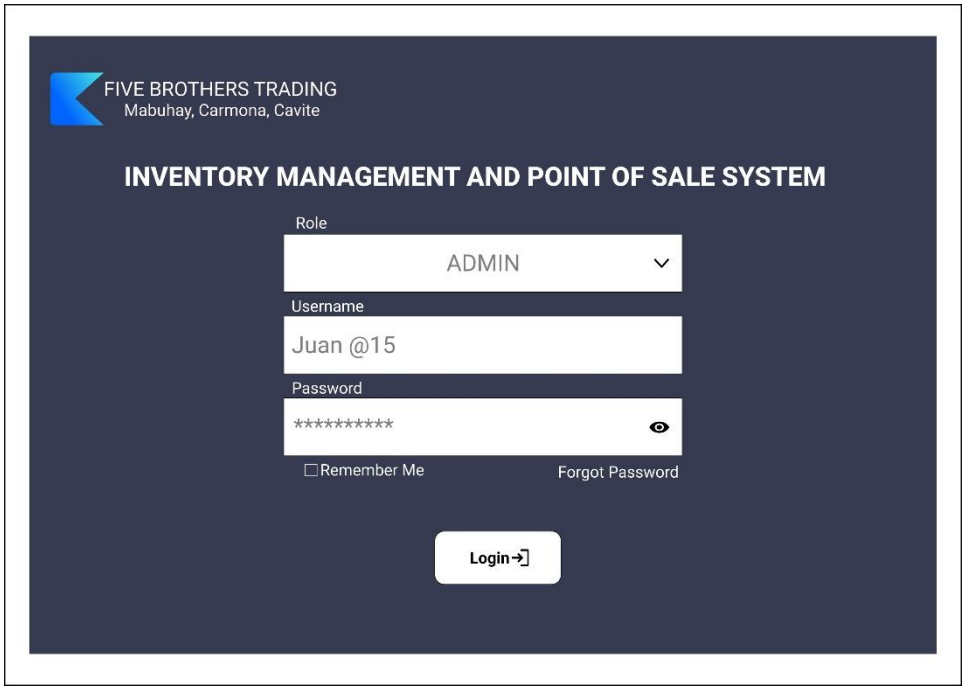


Figure 6. Screenshot of Login Page

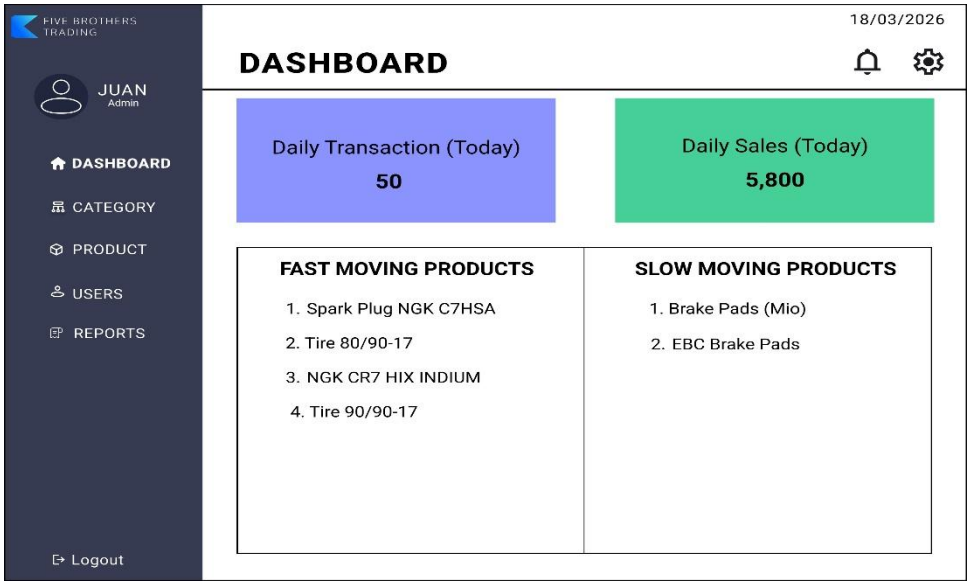


Figure 7. Screenshot of Dashboard

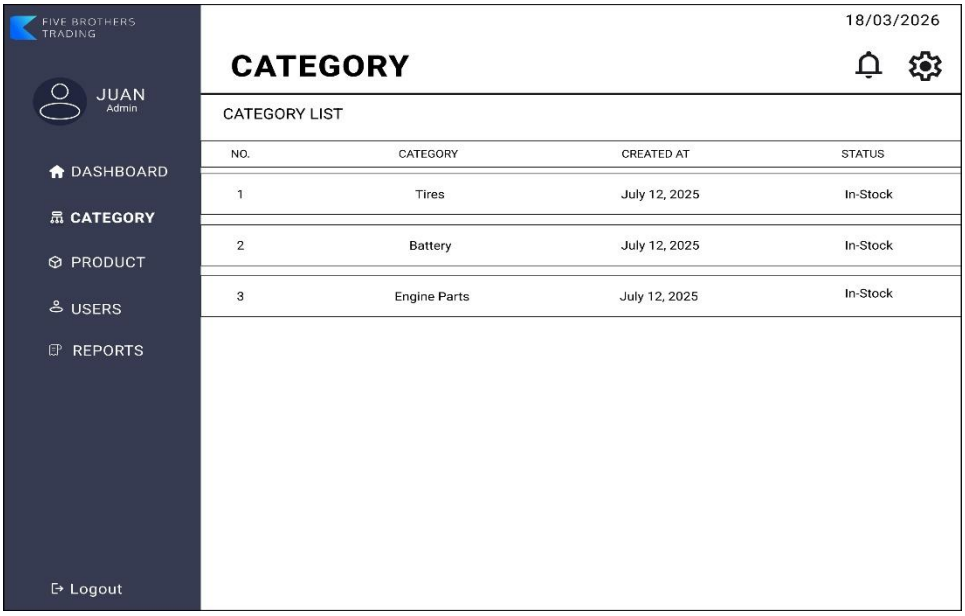


Figure 8. Screenshot of Category Page

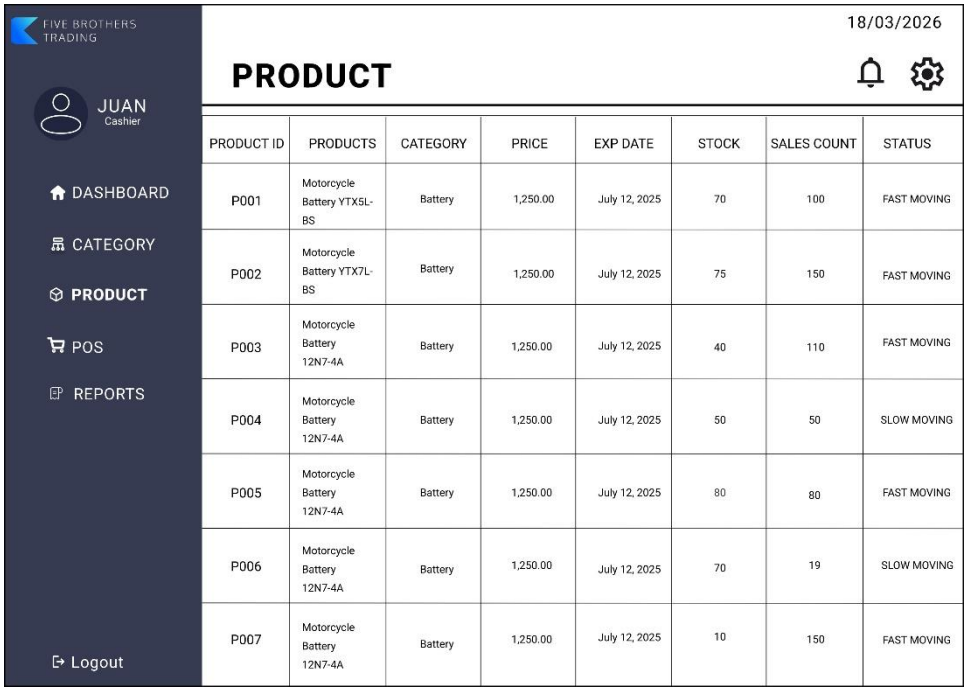


Figure 9. Screenshot of Products Page

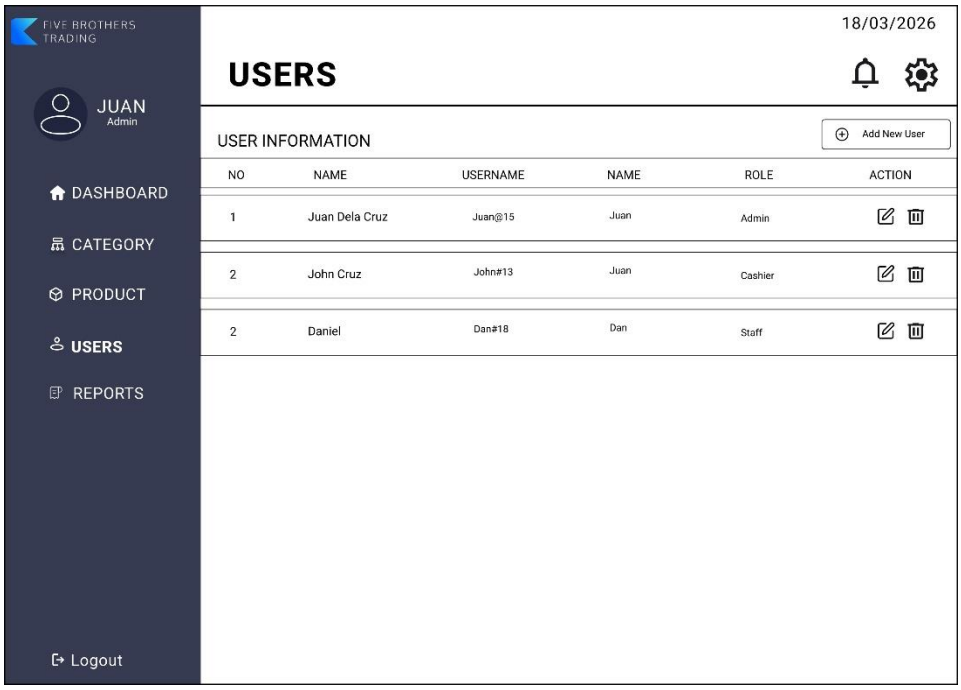


Figure 10. Screenshot of User Information

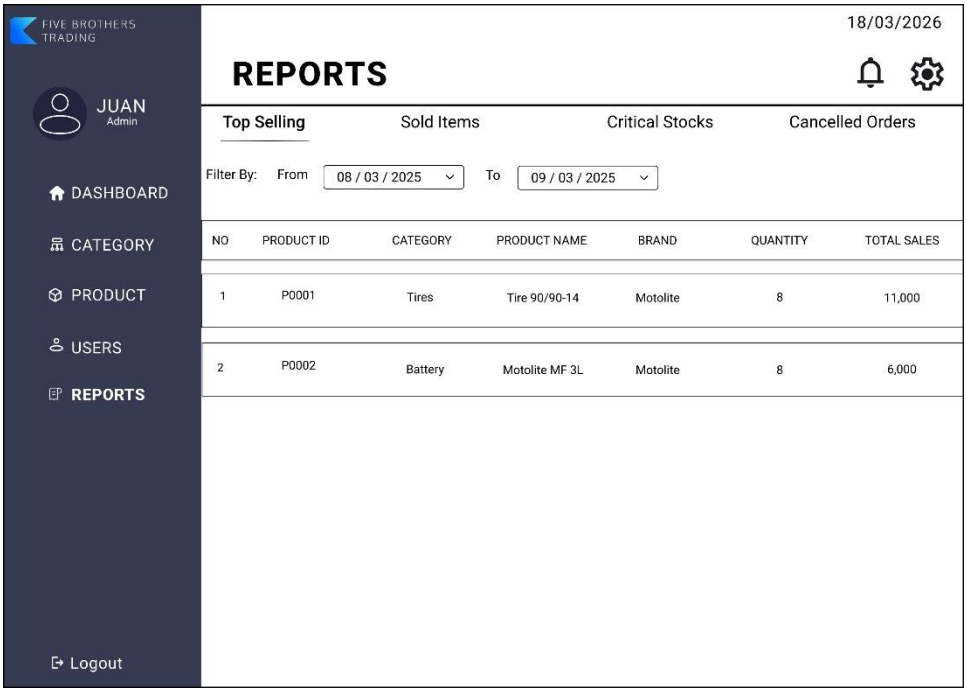


Figure 11. Screenshot of Records Top Selling Page

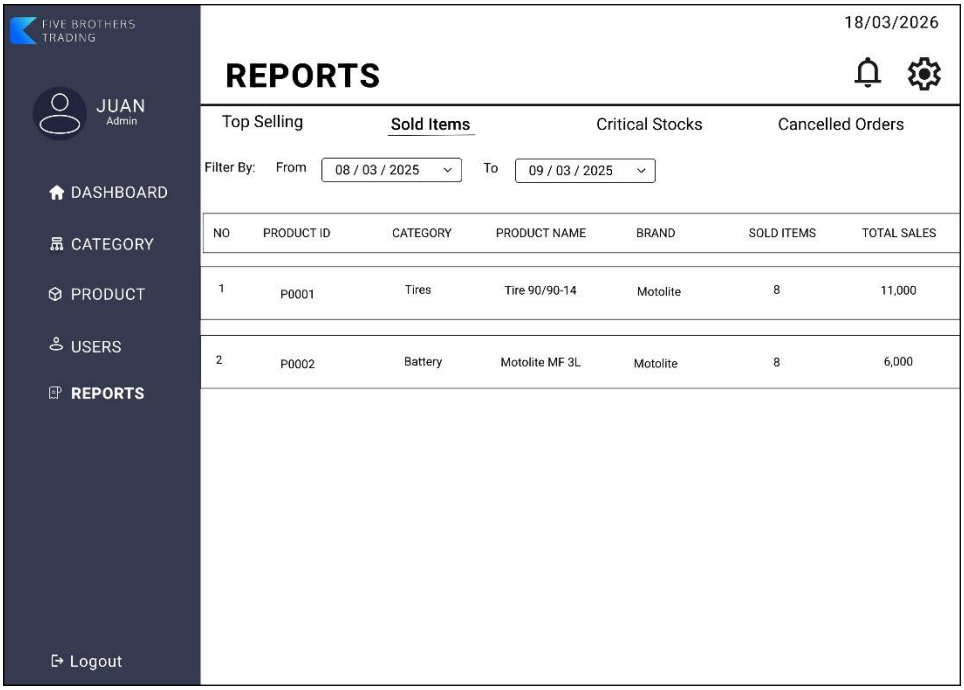


Figure 12. Screenshot of Records Sold Items Page

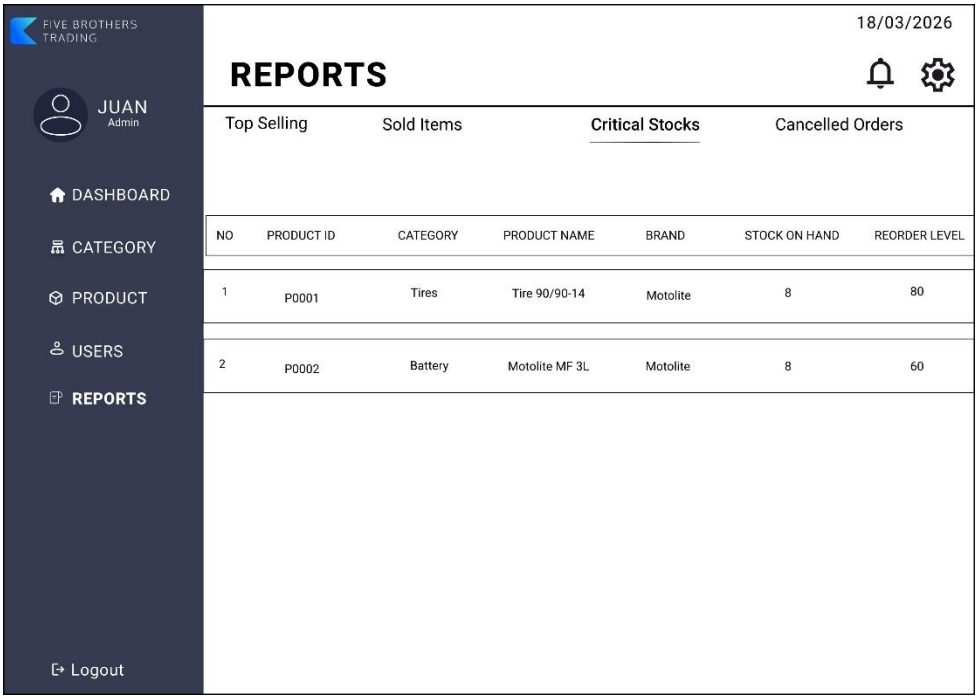


Figure 13. Screenshot of Records Critical Stocks Page

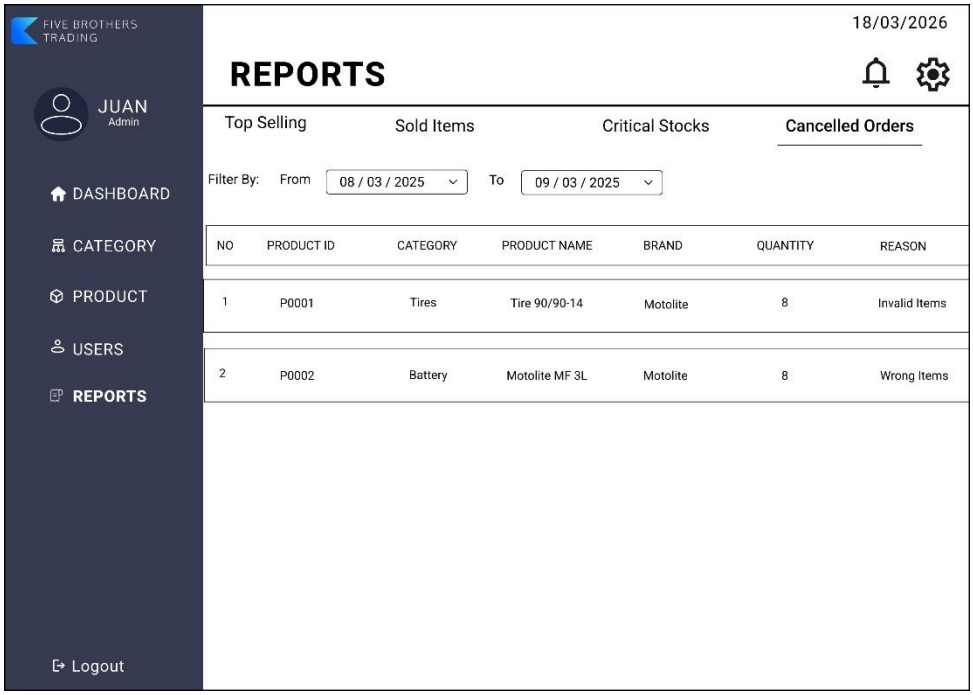


Figure 14. Screenshot of Records Cancelled Orders Page

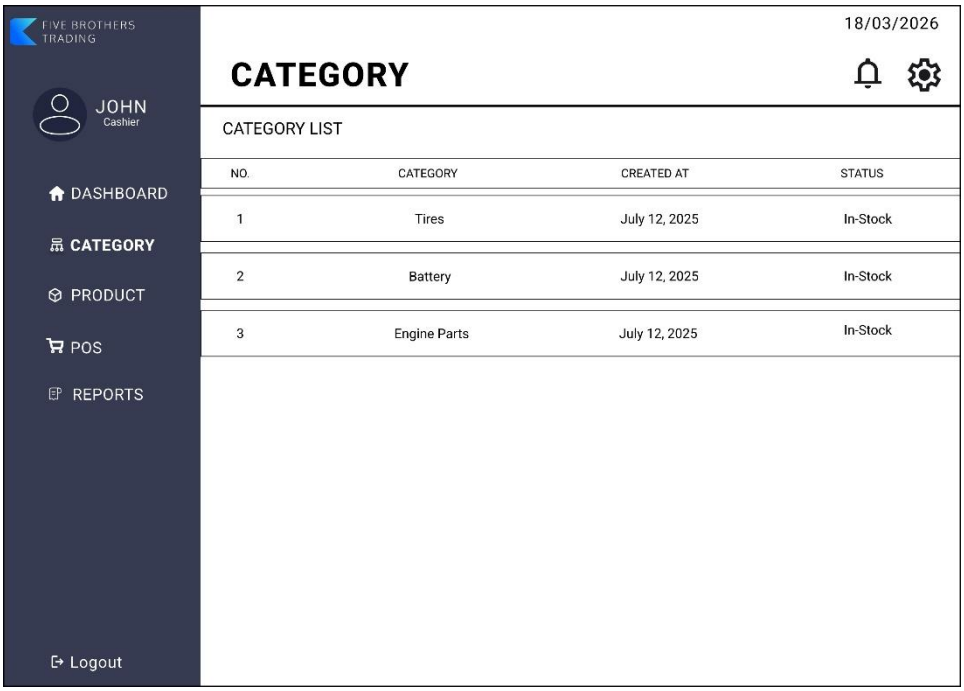


Figure 15. Screenshot of Category for Cashier Page

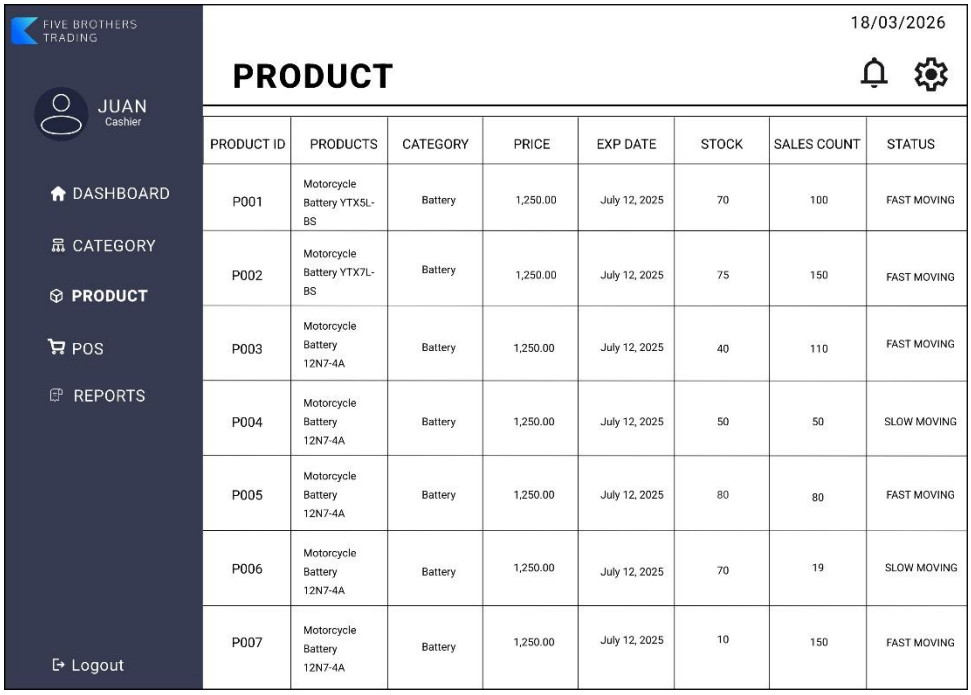


Figure 16. Screenshot of Products for Cashier Page

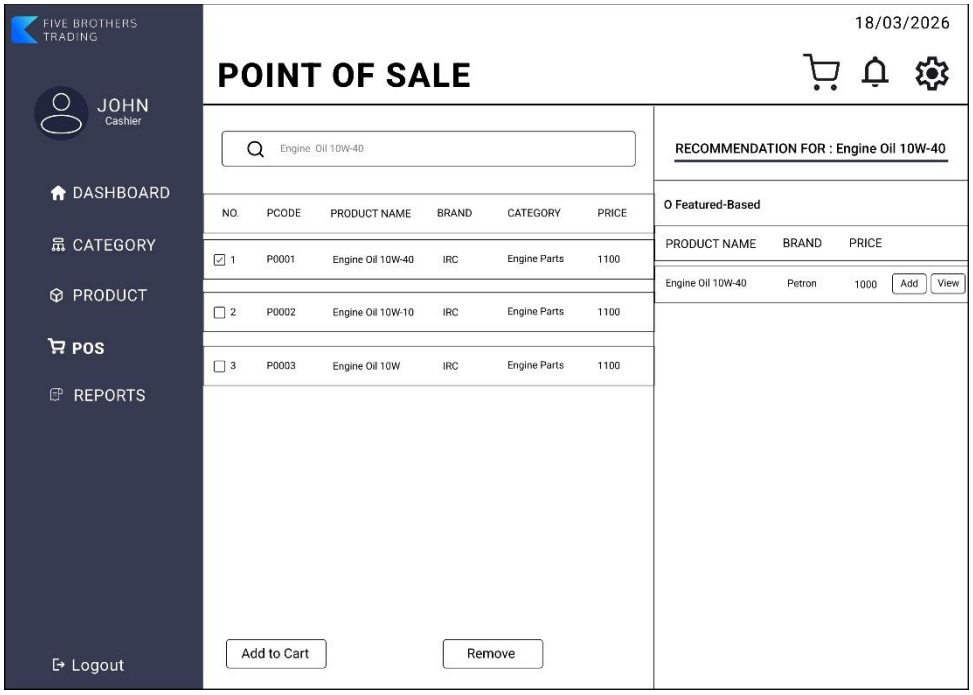


Figure 17. Screenshot of Point of Sale Part 1 for Cashier Page

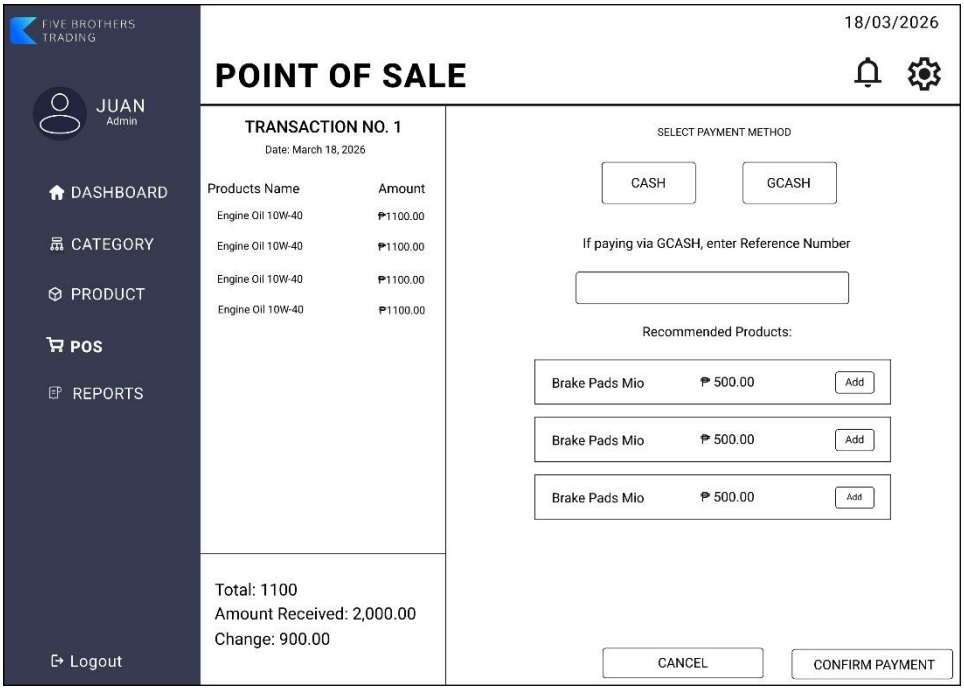


Figure 18. Screenshot of Point of Sale Part 2 for Cashier

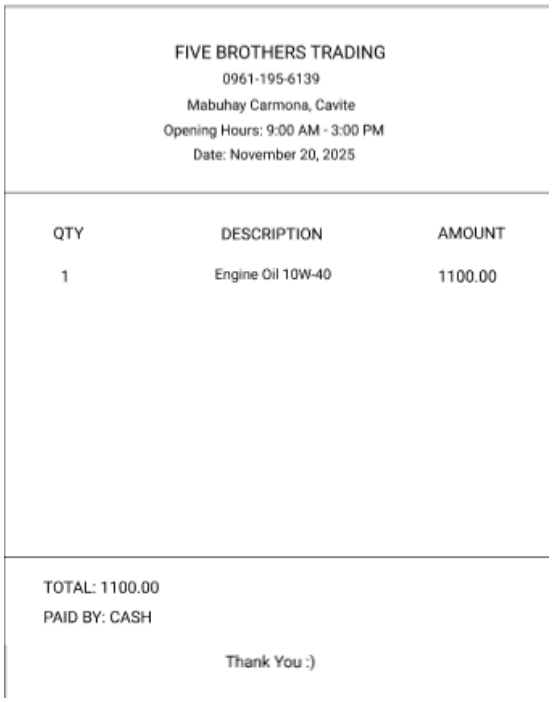


Figure 19. Screenshot of Receipt

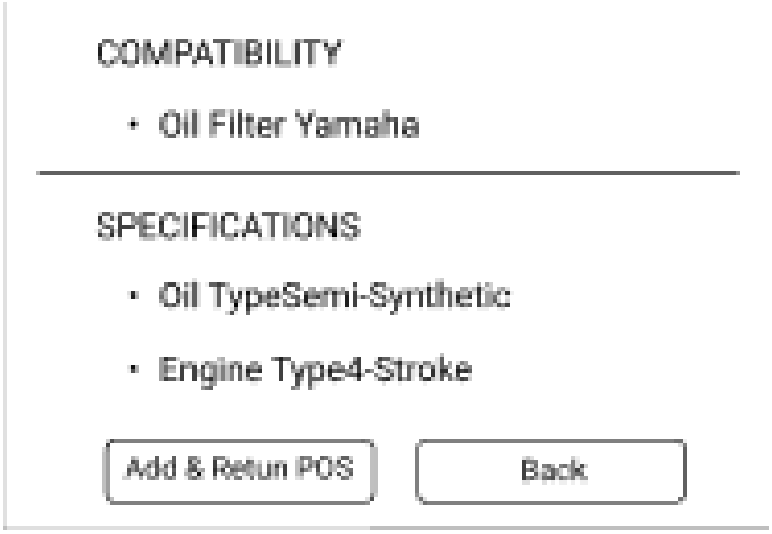
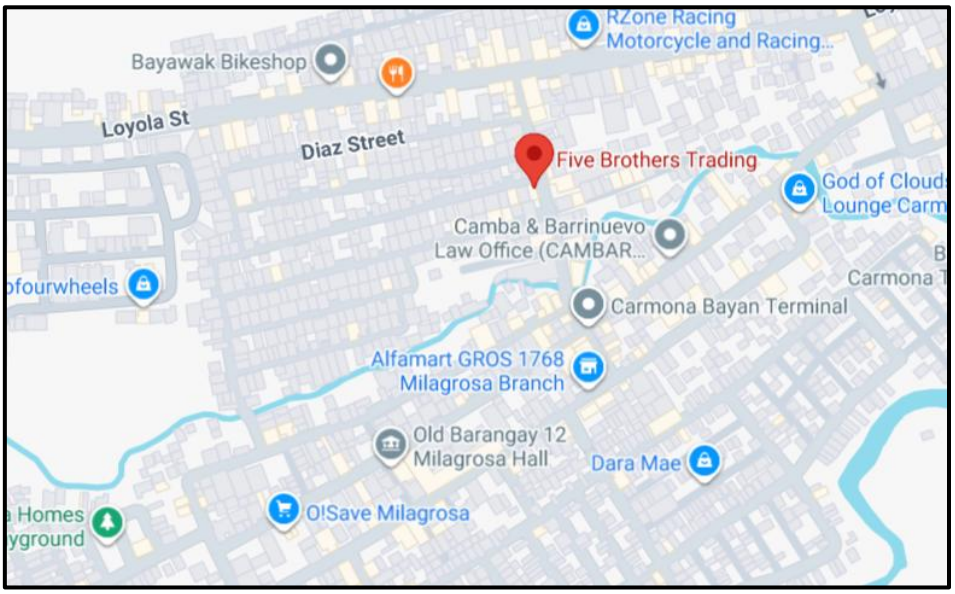


Figure 20. Screenshot of Recommendation of the System

LIST OF APPENDIX FIGURES

Appendix Figure 1. Five Brothers Trading Establishment



Appendix Figure 2. Five Brothers Trading Location Map